

Question 5

→ IPO & PAC chart

Input	Process	OUTPUT
- Zone: $A \leftarrow 20$ $B \leftarrow 40$ $C \leftarrow 15$ - Car, Bike, Van $\leftarrow 0$ - Check emergency - Check: - Zone - Vehicle - Permit	- Set all zones capacity - Validate each zone, vehicle, permit - Check capacity before entering - Check emergency → IF zone C & van → space + 2 rest space + 1 → IF space is avl bl then update counters else reject → update all counters	- Summary: - Total zone parked in - Total Cars - Total Bike - Total Van - Capacity of all zones - IF rejected THEN reason



Date:

Pseudocode

Cap-A $\leftarrow$ 20
Cap-B $\leftarrow$ 40
Cap-C $\leftarrow$ 15

Occ-A , Occ-B , Occ-C $\leftarrow$ 0
Car , Van , Bike $\leftarrow$ 0
rejected $\leftarrow$ 0

Repeat

INPUT "Enter numbers of expected vehicles", Total-vehicle
IF Total-vehicle $>$ 0
 OUTPUT "Invalid - Enter again"
ENDIF

Until (Total - vehicle $\geq$ 0)

For i = 1 TO Total-vehicle

OUTPUT "Vehicle", i

REPEAT

INPUT "Enter vehicle type (C/V/B)", type

 IF type $\neq$ "C" OR "B" OR "V" THEN

OUTPUT "Invalid"

ENDIF

UNTIL (type = ("C" OR "B" OR "V"))

REPEAT

INPUT "Enter category", C

IF C != 'F' OR 'S' OR 'G' THEN

OUTPUT "Invalid"

ENDIF

UNTIL (C = 'F' OR 'S' OR 'G')

REPEAT

INPUT "Has valid Permit (Y/N)", Permit

IF Permit != "Y" OR "N" THEN

OUTPUT "Invalid"

ENDIF

UNTIL (Permit = "Y" OR "N")

emergency = "N"

IF Permit = "N"

Repeat

INPUT "emergency vehicle (Y/N)", emergency

UNTIL (emergency != "N" OR "Y")

assigned ← FALSE

zone ← NULL

reason ← ""

IF Permit = "N" AND emergency = "N" THEN

reason ← "Invalid permit"

ELSEIF C = "F" THEN

space $\leftarrow$ 1

IF OCC - A + space $\leq$ CAP - A THEN

zone $\leftarrow$ "A"

OCC - A $+=$ 1

assigned $\leftarrow$ TRUE

Else

reason $\leftarrow$ "no space in Zone A"

ENDIF

ELSE IF C = "S" THEN

IF type = "V" THEN

space $\leftarrow$ 2

IF OCC - C + space $\leq$ Cap - C

OCC - C $+=$ 2

assigned = TRUE

ELSE

reason $\leftarrow$ "zone C is full"

ENDIF

ELSE

space $\leftarrow$ 1

IF OCC - B + space $\leq$ Cap - B THEN

zone $\leftarrow$ "B"

OCC - B $+=$ 1

assigned = TRUE

ELSE

reason $\leftarrow$ "zone B is full"

ENDIF

ENDIF

Date: _____

ELSE

IF type = "V" THEN
Space = 2

ELSE

Space = 1

ENDIF

IF OCC-C + space $\leq$ Cap-C THENZone $\leftarrow$ "C"OCC-C $\leftarrow$ OCC-C + space

assigned = TRUE

ELSE

reason $\leftarrow$ "Zone C is full"

ENDIF

ENDIF

ENDIF

IF assigned = TRUE THEN

IF zone = "A" THEN

remain $\leftarrow$ Cap-A - OCC-A

ELSE IF zone = "B" THEN

remain $\leftarrow$ Cap-B - OCC-B

ELSE

remain $\leftarrow$ CapC - OCC-C

ENDIF

OUTPUT "vehicle parked in zone: ", zone

OUTPUT "Remaining capacity", remain


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IF type "C" THEN
    car += 1
ElseIF type "B" THEN
    Bike += 1
ELSE
    Van += 1

ELSE
    OUTPUT "Vehicle rejected . Reason: ", reason
    rejected += 1

ENDIF
ENDFOR

OUTPUT " Summary"
OUTPUT " total vehicles: ", total
OUTPUT " Cars parked: ", car
OUTPUT " Bike parked: ", bike
OUTPUT " Van parked: ", van
OUTPUT " rejected count: ", rejected
OUTPUT " zone A occupied: ", OCC-A
OUTPUT " zone B occupied: ", OCC-B
OUTPUT " zone C occupied: ", OCC-C

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